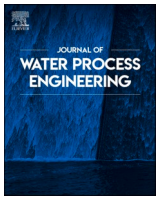
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Micro-aeration based anaerobic digestion for food waste treatment: A review

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ABSTRACT

In recent years, micro-aeration based anaerobic digestion technology has been utilized for food waste biodegradation because it can facilitate traditional anaerobic digestion technology to obtain higher food waste degrading efficiency. This paper aims to review previous studies on food waste micro-aeration digestion to enrich theories and provide bioreactor parameter guidance. Firstly, the underlying mechanism behind food waste biodegradation pathway is set forth to lay foundation for following micro-aeration researches. Subsequently, advantages and disadvantages of three oxygen addition strategies have been concluded, which can provide a guidance for food waste micro-aeration digestion bioreactor optimization. Afterwards, preponderant effects of micro-aeration technology on food waste hydrolysis, acidogenesis and methanogenesis are elaborated to supply solution. However, modified anaerobic digestion kinetic models under micro-aeration condition and optimized micro-aeration strategy for long-term bioreactor operation deserve further exploration. Therefore, future researches are needed to develop micro-aeration kinetic models and optimize steady-running micro-aeration bioreactor for food waste biodegradation.

1. Introduction

The amount of food waste (FW) originated from food processing and catering industry is increasing yearly. Approximately 931 million tons of FW were produced worldwide annually while more than 121 million tons were generated in China [1,2]. Therefore, it needs to be appropriately disposed to bring down its potential environmental hazard to the minimum level. On the other hand, FW is also a typical low-cost and high-energy biomass. Considering their negative environmental effects of tradition technologies for FW disposal such as landfill, incineration and composting, anaerobic digestion (AD) is a promising renewable technology for FW biodegradation because FW owns favorable characteristic such as large quantity and high bioavailability [3–5]. Meanwhile, in anaerobic bioreactor for FW disposal, anaerobic microorganisms can convert a wide range of complex organic waste into biogas including methane, which can be reprocessed to heat and electricity [6]. Therefore, FW can be considered as an ideal and promising substrate in anaerobic bioreactor and many researchers focused on its high methane production potential [7–9]. The highest methane potential of different types of FW (lipidic-rich FW, protein-rich FW and carbohydrate-rich FW) is in the range of 0.3 to 1.1 L CH_4/g VS which is apparently higher than traditional substrates such as lignocellulosic biomass and excess sludge [10,11]. Among diverse FW, lipid-rich FW containing fat, oil and grease (FOG) has the highest methane yield potential theoretically for 1.1 L CH_4/g VS in AD system which deserves wide application.

Though AD shows unique superiority in FW management, it also confronts with the following challenges including volatile fatty acid (VFA) and ammonia accumulation, instability, foaming and acidification [12,13]. In particular, lipid-rich FW is prone to contribute to AD process failure due to harmful intermediate compound such as long chain fatty acids (LCFA) accumulation. During lipid-rich FW biodegradation, FOG are subjected to lipolysis process to decompose into glycerol and LCFAs. Therefore, anaerobic bioreactor for single FW disposal has to be operated at low organic loading rates (OLR) [14]. Consequently, researchers have explored countermeasure to conduct FW biodegradation at high OLR in AD reactor including adding metal cations, natural absorbents and co-digestion substrate [15–17]. However, the insoluble compounds formed by metal cations can decompose in thermophilic AD bioreactor, and some metal cations can adhere to microbial

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cytomembrane which can impede FW conversion to biogas [17,18]. On the other hand, natural adsorbent completed by adsorption needs following treatment [15]. Therefore, it is meaningful to explore other appropriate methods to promote anaerobic FW biodegradation.

Recently, a burgeoning technology (micro-aeration treatment technology) has stepped into researchers' horizon for their positive effect on promoting the performance of anaerobic bioreactor. This technology can manifest in many aspects including augmenting functional microbial abundance and converting more types of substrates such as LCFAs for methane-producing archaea (MPA) growth [19–21]. Simultaneously, wipe off hydrogen sulfide to purify methane and promote volatile fatty acids (VFAs) decomposing efficiency are also positive effects of micro-aeration environment [22]. Based on the above-mentioned preponderance, a large number of researchers have concluded positive performance and future challenge of micro-aeration system to provide a guidance for subsequent research [22,23]. Therefore, traditional anaerobic digestion model No.1 (ADM1) has been promoted into ADM1-Ox to predict experimental methane production potential under micro-aeration conditions [11,24]. As for bioreactor for solid waste disposal, micro-aeration can promote hydrogen sulfide removal from biogas [25]. As regards to FW biodegradation, micro-aeration system shows great potential on enhancing FW hydrolysis efficiency in anaerobic bioreactor [26]. Micro-aeration system can contribute to a suitable cultivating environment for facultative acidogens including phylum *Firmicutes* and *Proteobacteria* and high amounts of extracellular enzymes which are beneficial to FW biodegradation [27]. Moreover, micro-aeration condition has been proved to be feasible in reliving LCFA accumulation which provides a novel horizon for lipid-rich FW management in AD bioreactor [28]. In addition, micro-aeration technology for FW treatment has been proved to promote energy self-sufficiency and economic benefits in particular cities such as Hongkong in China and Grüneck in Germany [29,30]. Hence, micro-aeration based AD system provides a feasible orientation to FW biodegradation which deserves deep-going investigation. However, there is still a lack of systematical summary about FW treatment under micro-aeration conditions due to the related studies remain deficient.

Therefore, the purpose of this paper is to sum up micro-aeration based AD for FW treatment according to previous studies. Firstly, definition and mechanism behind FW biodegradation under micro-aeration based AD condition especially electron pathway and ORP transformation have been described. Subsequently, essential considerations of micro-aeration intensity and duration have been discussed, especially oxygen or air addition strategies to create micro-aeration condition. Afterwards, positive effects of FW micro-aeration system have been summarized. At last, future perspectives and research gaps are put forward due to insufficient researches about micro-aeration kinetic models and bioreactor optimization.

2. Micro-aeration based AD for FW biodegradation: definition and mechanisms

2.1. Definition of micro-aeration based AD system

As a prosperous technology for organic waste biodegradation, micro-aeration based AD system means injecting a small dosage of oxygen or air into traditional AD bioreactor [22,23]. Furthermore, according to previous studies, relevant nouns such as limited aeration, micro-oxygenation, oxygenation, microaerobic and microaerophilic are also possible explanations to replace micro-aeration [22]. Though micro-aeration system represents that dissolved oxygen ranges from 0.1 to 1.0 mg/L in the bioreactor, dissolved oxygen in bioreactor is tough to be definitely expressed due to persist biological reaction in micro-aeration reactor [23]. Therefore, the oxidation-reduction potential (ORP) value is introduced to standardize micro-aeration system definition, and ORP value of micro-aeration system is in the range of −300 mV to 0 mV when determined by a standard hydrogen electrode [22]. The ORP value can indicate the net worth of intracellular metabolism dominated by electron transfer and redox balance which is a significant index in biologic systems [31]. Therefore, ORP value can precisely embody tiny transformation in oxygen concentration in liquid phase to reflect micro-aeration degree.

In general, ORP value ranging from −215 mV to −150 mV are universal for bioreactors utilized to create micro-aeration environment and thus numerous researchers have investigated effects of ORP value on the performance of micro-aeration bioreactor [32]. For example, when the ORP value ranges from −275 mV to −265 mV, hydrogen sulfide in the biogas can be removed [33]. In addition, the production of butanol and hydrogen can be promoted at a feasible ORP value in an electro-fermentation system [34]. Therefore, based on the purpose of diverse outstanding performance manifested in Fig. 1, micro-aeration rate and ORP value can be regulated for organic waste treatment under micro-aeration condition.

In FW micro-aeration based AD system, previous studies have proved that the optimal ORP level for FW acidogenic fermentation process is in the range of −100 mV to −200 mV. Furthermore, for the sake of avoiding LCFAs accumulation during lipidic-rich FW treatment, the ORP value in anaerobic bioreactor should be maintained at −255 ± 47 mV which is suitable for the growth of LCFA-degrading facultative bacteria such as genus *Pseudomonas* [28]. Therefore, ORP value is one of the most momentous indexes in FW micro-aeration based AD system which can be regulated to meet the requirements of various research goals.

2.2. Mechanisms of micro-aeration based AD for FW biodegradation

It is well-known that traditional FW biodegradation pathway in AD system is a complex multi-step process. In general, this process contains four steps including hydrolysis, acidogenesis, acetogenesis and methanogenesis [35]. Therefore, AD process experiences a lengthy and intricated duration, and harmful intermediate compounds such as hydrolysate can accumulate in AD bioreactor which will cause ineffective methane production, reactor instability, sludge floating and foaming [36,37]. Furthermore, hydrolysis has been recognized as the rate-limiting step in AD process because it involves complicated reaction by diverse enzymes. In general, these enzymes can convert insoluble polymers such as carbohydrates, proteins and lipids into soluble monomers such as monosaccharides, amino acids and LCFAs [38]. Meanwhile, at the initial stage of AD system, complicated organic compounds in FW can rapidly convert into VFAs, which will contribute to reactor acidification due to low content of buffering agent [39,40]. Therefore, hydrolysate such as LCFAs and free ammonia are two common inhibitors in traditional AD process and their inhibition concentration on AD system has been investigated in previous studies [41–43]. Furthermore, anaerobic microorganism needs rigorous growing condition and a long generation duration, especially MPA. Therefore, if FW is recognized as the single substrate for AD process, it should be maintained at a low organic loading rate (OLR) of 1.5 g-VS/L d in mesophilic anaerobic bioreactor [44]. Additionally, FW can also co-digest with other solid waste such as lignocellulosic waste or blend with micronutrients because they can provide efficient trace elements and buffer agent for AD process at high OLR [45,46].

Micro-aeration based AD system shows great potential in alleviating LCFAs accumulation and applying more types of organic compounds as substrate for MPA growth. Therefore, it can be applied to alleviate above-mentioned problems because it provides a peculiar environment that situates between anaerobic and aerobic conditions which supplies favorable niches for both anaerobic and aerobic bacteria [20,28]. In micro-aeration system, the small dosage of oxygen can be recognized as the terminal electron acceptor for organic waste biodegradation, which replenish the lack of an effective mechanism to transport electrons in traditional AD process [20,22]. In addition, acetyl-CoA produced in AD process undergoes tricarboxylic acid cycle in aerobic process to be completely oxidized into carbon dioxide in aeration system [20].

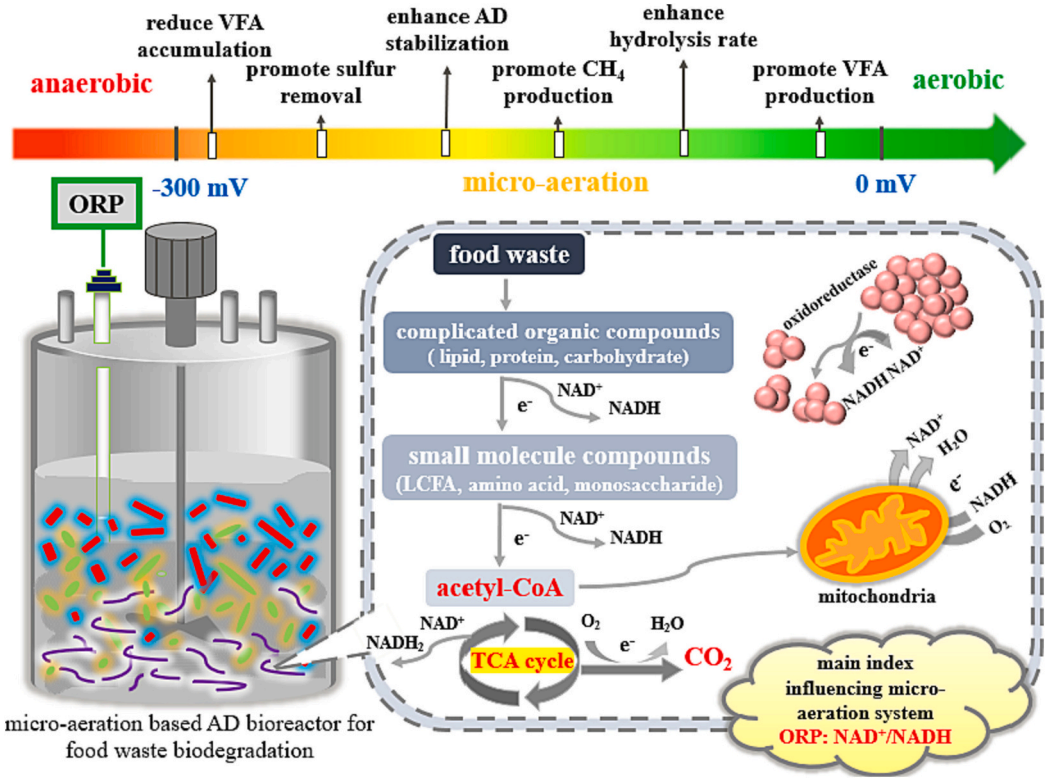


Fig. 1. Relationship between ORP and electron transfer mechanism and purposed ORP value for diverse research targets in micro-aeration based AD bioreactor for FW treatment.

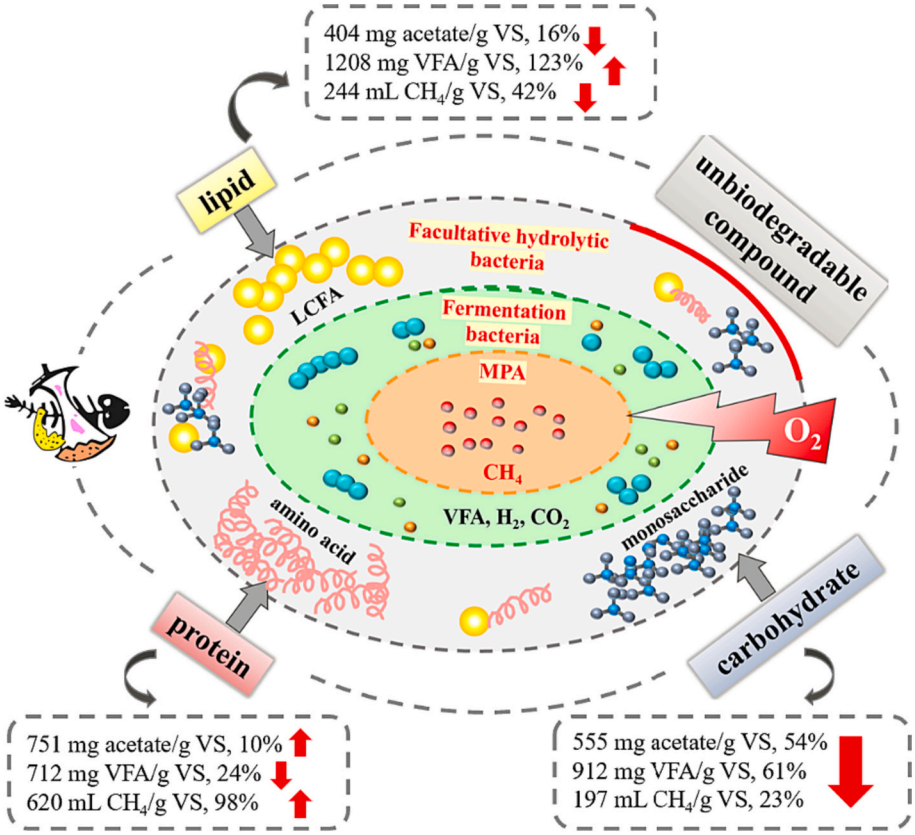


Fig. 2. Microorganism flocs and their function under micro-aeration based AD system for different composition of FW treatment (calculated by data which was carried out by 120 L-air pretreatment [54], the red arrow refers to the increase or decrease percentage compared with traditional AD). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Meanwhile, oxygen contributes to nicotinamide adenine dinucleotide (NAD^+) conversion into NADH [47]. NAD^+ should be regenerated from NADH through the production of fermentation intermediates such as VFAs and lactate. NADH oxidation process produces a good deal of energy to promote adenosine triphosphate synthesis and then promotes hydrolysis acidification efficiency [47]. Fig. 1 presents the electron transfer mechanism in micro-aeration based AD system for FW biodegradation. Generally, micro-aeration based AD system provides positive environment for energetic conversion of intermediates such as acetyl-CoA to maintain stability of AD process, and thus it can be recognized as a promising technology for complicated FW treatment.

It is well-acknowledged that ORP value is the most precise indicator to describe oxygen concentration in micro-aeration condition because ORP value presents linearly variation tendency with the logarithmic of system oxygen concentration. Nevertheless, oxygen solubility in aqueous phase can be readily affected by pH and temperature in external environment. In FW micro-aeration based AD system, dissolved oxygen contributes to redox pairs formation in cytoplasm microcosmically. Among various redox pairs, $NADH/NAD^+$ and $NADPH/NADP^+$ (nicotinamide adenine dinucleotide phosphate) plays uppermost roles in intracellular metabolism. Therefore, due to redistribution of metabolic flux and improvement of fermentation product yield, ORP value can be altered and thus bioreactor performance can be transformed. As depicted in Fig. 1, dominant positive effects of micro-aeration system transform from the minimum aeration degree to the maximum aeration degree including taking the edge off VFAs accumulation, facilitating sulfur removal, enhancing methane production, promoting hydrolysis efficiency and heightening VFAs production [22].

Microscopically, facultative microbes consume and convert oxygen molecule to generate reactive oxygen species. As described in Fig. 2, diverse microbial species tend to form flocs to stand up to adverse external conditions including reactive oxygen species in micro-aeration based AD process. In these flocs, facultative hydrolytic bacteria are in the outmost layer and wrap fermentative bacteria. These facultative bacteria convert added oxygen into reactive oxygen species, which can decompose bioavailable compounds into various small molecular intermediates for fermentative bacteria utilization. Subsequently, fermentative bacteria surround around MPA to serve substrate such as acetate and carbon dioxide to carry out methanogenesis process. Therefore, facultative bacteria possibly play the role as shield to protect fermentative bacteria and MPA, and maintain their liveness and vitality under micro-aeration condition. As for ORP value that reflects the residual oxygen concentration, it can descend from the outmost layer (hydrolytic bacteria) to the innermost layer (MPA). Therefore, in the wake of ORP transformation value in FW bioreactor, the abundance of functional and dominant microorganism alters. Generally, aeration influenced the abundance, evenness and structure of the bacterial and MPA species in the long term [48]. For example, micro-aeration results in MPA abundance variation such as *Methanosarcina* decrease and *Methanoculleus* increase [49]. Moreover, methanogenic activity of MPA in micro-aeration based AD system is higher than that in traditional anaerobic bioreactor under higher sulfide concentrations [50]. As for FW biodegradation, previous studies have demonstrated that bacterial community in micro-aeration based AD system surpasses that in single anaerobic acidogenic fermentation and aerobic system [27,51]. Especially facultative bacteria *Clostridium*, *Oscillospira*, *Anaerofilum*, *Pseudomonas* and *Ruminococcus* in phylum *Firmicutes* have taken account of 43.7 % of the whole microbial species which accelerates FW hydrolysis process in acidogenic bioreactor [52]. In addition, another study indicates that 58 % *Firmicutes* and 42 % *Proteobacteria* accounts for the phylum in anaerobic bioreactor, while micro-aeration FW treatment bioreactor contains 72 % *Firmicutes*, 2 % *Proteobacteria*, 24 % *Bacteroidetes* and 2 % *Actinobacteria* [27]. Moreover, another micro-aeration method (thermophilic micro-aerobic pretreatment) also augment phylum *Firmicutes* abundance to produce extracellular enzymes [53]. This indicates that micro-aeration based AD process contributes to the development of a more diverse bacterial population especially facultative hydrolysis bacteria in phylum *Firmicutes* to metabolize FW than traditional AD process.

3. Operational strategies to create micro-aeration condition

The scheme of oxygen dosage requires systematic investigation because oxygen dosage amount influences system ORP value, which is the most essential indicator of micro-aeration bioreactor performance. In general, excess oxygen dosage can contribute to the generation of novel intermediate organic compounds and detriment the structure of MPA, and inadequate oxygen dosage shows an inconspicuous influence on micro-aeration bioreactor [55]. Therefore, operational strategy of appropriate oxygen addition is crucial in micro-aeration based AD system. Previous studies mainly focused on oxygen addition at different stages such as micro-aeration during pretreatment phase, intermittent micro-aeration, persistent micro-aeration during whole experiment as depicted in Fig. 3.

3.1. Adding oxygen during pretreatment phase

During pretreatment phase, oxygen supply strategy is single-pass added directly into the sludge or the substrate before the formal experiments. In general, due to the intrinsic shape of anaerobic bioreactor with superior air impermeability, most researchers add oxygen during pretreatment phase to improve micro-aeration bioreactor performance. This operation approach is simple and convenient on account of inferior requirements on persistent ORP regulation. As for micro-aeration strategies, several milliliters of oxygen have been adopted to create micro-aeration condition for FW biodegradation as shown in Table 1. As regards to bioreactor performance after micro-aeration, 5 mL oxygen/g-VS oxygen addition contributes to the maximum cumulative methane yield [53]. In addition, oxygen has low solubility in water media at about 0.21 mmol at 37 °C which erodes the significance of surplus oxygen existence and augments their interference on biogas composition [56]. As for 1 L micro-aeration bioreactor such as sequencing batch reactor (SBR), 15 mL oxygen can promote the largest increase in methane production with more than 10 % percentage compared with no aeration reactor [57,58]. The percentage is larger than that of 5,10,20 mL oxygen addition which manifest that 15 mL oxygen is the optimal one to create micro-aeration condition for 1 L bioreactor. Moreover, bioreactor with 15 mL oxygen addition also performs slight decrease of ammonia nitrogen removal and 10 % increase of COD removal at the same time [57].

Moreover, due to the diverse ingredient and physical characteristics of FW, the optimal micro-aeration intensity during FW pretreatment phase deserves profound consideration. Fig. 2 presents total VFA production, acetate production and methane production of single bioavailable substance in FW such as lipid, protein and carbohydrate according to previous studies with micro-aeration rate at 120 L-air/d [54]. It is apparent that lipid-rich FW shows inferior bioavailability than protein-rich and carbohydrate-rich FW with micro-aerated pretreatment. Nevertheless, substrate tank can be open to air without deliberate micro-aeration to enrich LCFAs-biodegrade facultative microbes which provides a novel horizon for lipidic-rich FW micro-aeration treatment [28]. Therefore, FW with different contents of bioavailable substance can be pretreated by appropriate micro-aeration strategy to regulate bioreactor performance.

In conclusion, micro-aeration during pretreatment phase is an effective method for FW biodegradation. However, FW characteristic and micro-aeration intensity should be taken into sufficient consideration to maintain at opportune range and thus micro-aeration based AD technology can play a tremendous role to facilitate FW biodegradation and bioreactor performance.

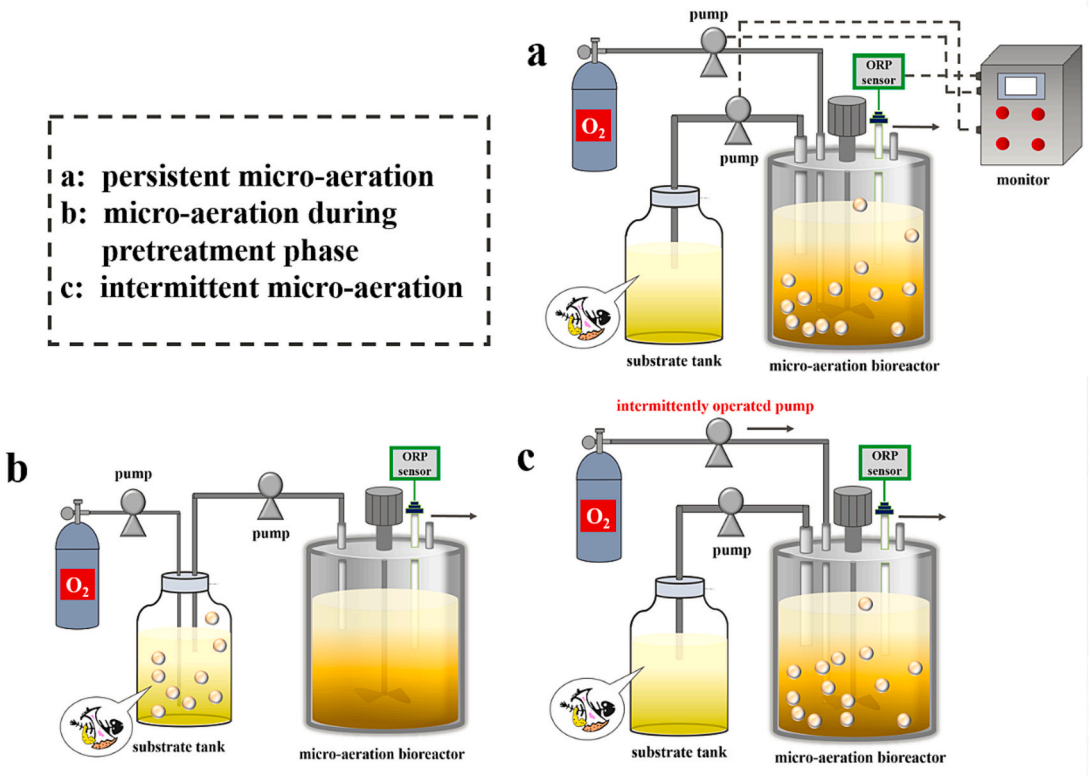


Fig. 3. Diagram of three oxygen addition strategies.

3.2. Adding oxygen with intermittent micro-aeration

During the pretreatment phase, oxidizing gas such as air or oxygen can be flushed into anaerobic bioreactor to regulate system ORP into satisfactory value through aeration head. Nevertheless, with the persistent operation of biological reaction, oxygen is gradually utilized by facultative microorganisms until it is completely consumed. Therefore, in order to maintain sufficient water-soluble oxygen in micro-aeration bioreactor, intermittent micro-aeration during the whole experiment phase is adopted to ensure appropriate oxygen content to regulate ORP value. Based on this, oxygen transfer rate and oxygen utilization rate should be concerned and pondered to determine applicable oxygenation intervals. As shown in previous studies, intermittent micro-aeration at a rate of 3.0 mL O_2/L_R-d for 5 min every 2 h can promote the highest cellulose hydrolysis efficiency compared with other intermittent micro-aeration schemes [59]. Meanwhile, air can be micro-aerated at a rate of 3.2 to 30 mL-air/g-TS d during 15 seconds per minute to promote VFAs production [22].

As regards to intermittent micro-aeration strategy for FW treatment, leach bed reactor can be micro-aerated at four frequencies including aeration for 12 min every 6 h, 3 h, 2 h and no aeration with a flow rate of 1.0 L-air/min. The optimum micro-aeration intensity is 258 mL-air/g-TS d because it can accelerate substance decomposition efficiency by 5 % and enhance methane production by 18 %. In addition, the generation of acetate and butyrate is more than three times higher than no micro-aeration condition and thus this optimum micro-aeration frequency can contribute to the maximum methane yield of 0.27 L CH_4/g-VS [26]. Based on this research, a packed bed reactor can be micro-aerated at a frequency of 15 minutes every 2 h per day to create the similar micro-aeration intensity of 254 mL-air/g-TS d, which contribute to 250 % increase of VFA production and 200 % increase of substrate decomposition efficiency compared with the bioreactor without micro-aeration. Therefore, 258 mL-air/g-TS d has been recognized as an acceptable micro-aeration intensity to facilitate the performance of micro-aeration bioreactor for FW biodegradation. Nevertheless, intermittent micro-aeration operation strategy is determined by the flow rate of the air pump. Another intermittent micro-aeration dosage is based on bioreactor volume. When aeration dosage has been controlled under specific aeration of 37.5 mL-O_2/L_R-d according to previous studies, cumulative biogas can be increased by 23 % compared with the bioreactor without aeration [60].

In conclusion, intermittent oxygenation during the whole experiment phase has large scope of application for FW biodegradation compared with single micro-aeration during pretreatment phase. Meanwhile, 258 mL-air/g-TS d total micro-aeration intensity has been proposed to serve as a guidance for future intermittent micro-aeration strategy. Therefore, intermittent micro-aeration is a more potential micro-aeration strategy.

3.3. Adding oxygen with persistent micro-aeration

Persistent micro-aeration during the whole experiment phase can be considered as the ideal micro-aeration strategy. However, micro-aeration intensity must be strictly monitored to maintain the normal vital activity of MPA. According to previous researches, bioreactor can be equipped with a pH sensor and an ORP sensor to automatically monitor system pH, temperature and aeration frequency. The modified micro-aeration bioreactor can experience long-term operation to maintain appropriate ORP scope and enrich facultative microorganisms at selective ORP value. Therefore, persistent micro-aeration strategy is convenient to operate due to automatic pH, temperature and aeration frequency regulation, the modified bioreactor performance is relatively stable as regards to food waste hydrolysis, acidogenesis and acetogenesis processes [61].

Previous studies have proved that Supervisory Control and Data Acquisition (SCADA) system can detailed describe and control the performance of micro-aeration bioreactor [62]. An electrically actuated ball valve can transform the oxygen through a diffuser from a pressurized oxygen cylinder, which can conduct automatic adjustment of opening size according to signal from the SCADA system to maintain the ORP

Table 1
Oxygen supply strategies under micro-aeration based AD system.

Micro-aeration strategy	Bioreactor type and working volume	Micro-aeration intensity	Bioreactor performance or their comparison with AD bioreactor	Reference
Intermittent micro-aeration for 12 min every 3 h in the bottom of LBR	4.6 L LBR-10 L UASB	258 mL-air/g-TS d	66.3 % hydrolysis degree, 79.6 % decomposition of carbohydrate, 69.7 % decomposition of protein, 1.05 g-COD/g-VS yield in LBR and 0.27 L CH_4/g-VS in UASB	[26]
Intermittent micro-aeration for 3 mL/min every day in the bottom of CSTR	3 L two-phase CSTR	From 5 mL-O_2/L_R-d to 7 mL-O_2/L_R-d	Slight decrease of pH, 10 % decrease of NH_3-N concentration, 8 % decrease of VFA, 1.5 % increase of SCOD	[27]
Intermittent micro-aeration for 15 min every 2 h in the bottom of PBR	1.5 L PBR and 9 L UASB	254 mL-air/g-TS d	Slight decrease of pH, 200 % increase of substrate decomposition, 250 % increase of VFA	[32]
Persistent micro-aeration	5 L gas circulation-type bioreactor	7.5 % (v/v) air of biogas-generation rate	Slight decrease of TOC, 99.9 % H_2S removal, 75 % decrease of VFA	[49]
Intermittent micro-aeration every 2 h based on ORP value from −200 to −100 mV	0.5 L serum vial bottle	274 mL-air/g-TS d	62.8 % removal of VS, 61.6 % removal of TCOD, 89.4 % decomposition of carbohydrate, 78.3 % decomposition of protein, 81.1 % decomposition of lipid, around 36.4 % increase of VFA	[51]
Pretreatment with 0, 24, 48 and 96 mL oxygen	1 L serum vial bottle	0, 5 mL, 10 mL and 20 mL-oxygen/g-VS	5 mL oxygen/g-VS creates the maximum cumulative methane yield	[53]
Pretreatment with 5 L/h air	1 L serum vail bottle	120 L-air/d	Lipid-rich FW: 14.5 % increase of VFA, 9 % decrease of methane, 8.7 % decrease of energy put; Carbohydrate-rich FW: 37.6 % decrease of VFA, 6 % increase of methane, 6.3 % increase of energy put; Protein-rich FW: 5 % increase of VFA, 45 % increase of methane, 45.8 % increase of energy put	[54]
Pretreatment with several milliliter air	1 L wide-mouth reagent bottle	5 mL, 10 mL, 15 mL and 20 mL air	5 mL O_2: 3 % increase of methane production 10 mL O_2: 2 % decrease of methane production 15 mL O_2: 22 % increase of methane production 20 mL O_2: 18 % increase of methane production	[58]
Pretreatment with several milliliter oxygen	1 L SBR	5 mL, 10 mL, 15 mL 20 mL O_2	5 mL O_2: 1 % decrease of methane production, slight increase of ammonia nitrogen removal, 23 % increase of COD removal; 10 mL O_2: 8 % increase of methane production, slight increase of ammonia nitrogen removal, 16 % decrease of COD removal; 15 mL O_2: 13 % increase of methane production, slight decrease of ammonia nitrogen removal, 10 % increase of COD removal; 20 mL O_2: 12 % increase of methane production, slight decrease of ammonia nitrogen removal, 28 % decrease of COD removal	[57]
Persistent micro-aeration	1.0–1.5 L CSTR	4.62, 7.69, 15.4 mL-air/L_R·min	4.62 mL/(L_R·min): 123 % increase of VFA production 7.69 mL/(L_R·min): 93 % increase of VFA production 15.4 mL/(L_R·min): 46 % increase of VFA production	[61]
Intermittent micro-aeration in the bottom of bioreactor	0.2 L serum vial bottle	37.5 mL-O_2/L_R-day for four days	23 % increase of methane production, 56 % increase of hydrolysate COD, 330 % increase of total VFA	[60]
Pretreatment with 400 L oxygen for 1 h	20 L semi-pilot scale AnMBR	400 L O_2	11 % increase of cumulative hydrogen production, 10 %	[67]

(continued on next page)

Table 1 (*continued*)

Micro-aeration strategy	Bioreactor type and working volume	Micro-aeration intensity	Bioreactor performance or their comparison with AD bioreactor	Reference
			increase of VFA production, around 25 % increase of substrate COD removal	

level in the range of −310 mV to −290 mV, and thus it can reduce hydrogen sulfide concentration in biogas [63]. In addition, another control system equipped with an S7-200 smart Programmable Logical Controller (PLC) is utilized to regulate micro-aeration intensity to enhance acetate production at ORP value lower than −200 mV [61]. In general, persistent micro-aeration controlled by ORP regulated oxygenation is feasible to optimize the performance of bioreactor.

Another persistent micro-aeration intensity depends on biogas generation rate. In general, gas-liquid flow state and hydraulic parameters can affect dissolved oxygen rate in micro-aeration bioreactor [64]. Therefore, appropriate ratio of micro-aeration rate per biogas generation rate is an essential index to control micro-aeration rate. According to previous studies, air can be injected into the bioreactor at a rate of approximately 7.5 % (v/v) of the biogas-generation rate to strengthen phylogenetic diversity of microorganisms in FW micro-aeration bioreactor [49]. However, this method cannot realize full utilization of oxygen though it proceeds continuous aeration and specific micro-aeration rate is tough to regulate. On the other hand, another micro-aeration strategy which involves oxygen recirculation can enhance the contact between oxygen, liquid phase and FW theoretically because it can increase the solubilization of oxygen in liquid phase and promote liquid-air, substrate-air and substrate-microorganism contact area [65]. This method has been applied to improve mass transfer during complex biological reaction and promote the removal efficiency of toxic compounds such as benzene and xylenes [66]. Nevertheless, oxygen recirculation has not been applied into FW micro-aeration system.

According to the above mentioned three operational strategies to create micro-aeration condition for food waste treatment, we can conclude several viewpoints as guidance for future relevant researches.

(1) As regards to lab-scale one phase micro-aeration bioreactor aimed at enhancing bioreactor stability, intermittent micro-aeration during the whole experiment can be considered as the optimal micro-aeration strategy to take abundant utilization of added oxygen and total aeration intensity can be controlled at 258 mL-air/g-TS d.
(2) As regards to lab-scale two phase micro-aeration bioreactor aimed at enhancing FW hydrolysis and acidogenesis fermentation, micro-aeration during pretreatment phase can be considered as the optimal micro-aeration strategy and the aeration intensity can be controlled at 5 mL-oxygen/g-VS for pretreatment bioreactor such as LBR and PBR.
(3) As regards to industrial-scale micro-aeration bioreactor for FW treatment, persistent micro-aeration during the whole experiment with automatic monitoring system can be considered as the optimal micro-aeration strategy on the basis of energy self-sufficiency and economic benefits.

4. Facilitation effects of micro-aeration on FW biodegradation

Based on the above-mentioned operational strategies, micro-aeration based AD system has been applied in FW biodegradation due to its facilitation effects on bioreactor. In conclusion, it can improve the performance of traditional anaerobic bioreactor in the following aspects such as strengthening FW hydrolysis and acidogenic fermentation, regulating biogas production and enriching microbial species.

4.1. Strengthening FW hydrolysis and acidogenic fermentation

In traditional AD reactor for FW biodegradation, hydrolysis and acidogenic fermentation involves the process of solubilizing complex organic compounds into small soluble compounds. Hydrolysis process requires the participation of diverse enzymes and it has been considered as the rate-limiting step [68,69]. Generally, FW with the characteristic of high lipid contents, low ratio of C/N further prolongs hydrolysis duration [70]. Furthermore, acidogenic fermentation is a dynamic course which contributes to the production of hydrogen and VFAs especially acetate for following procedure. Therefore, hydrolysis and acidogenic fermentation process undoubtedly lays the foundation for subsequent methanogenesis. Previous studies have focused on promoting hydrolysis and acidogenic efficiency including regulating pH, pretreating substrate, adding allogenic substances such as enzymes and microorganisms [71–73].

Nevertheless, micro-aeration based AD system can effectively promote FW hydrolysis and acidogenic fermentation efficiency [26,51,60]. The presence of oxygen impacts the hydrolysis rate and efficiency because it can enrich facultative hydrolytic and acidogenic microbes. In addition, hydrolysis reaction is driven by extracellular enzymes such as amylase, protease and lipase in micro-aeration based AD bioreactor, which promote the conversion rate of food waste into large number of acidogenic substances [68,74]. Hence, researchers take micro-aeration rate into consideration to explore the optimal ORP for FW hydrolysis and discover that the optimal ORP level for VFA production is in the range of −100 mV to −200 mV [51]. Moreover, 387 mL-air/g-TS d micro-aeration rate can increase more than 15 % FW hydrolysis degree compared with traditional AD condition in leach bed reactor [26]. This provides a guidance about micro-aeration intensity for future FW hydrolysis and acidogenic fermentation reactor.

Micro-aeration based AD system not only accelerates hydrolysis and acidogenic fermentation, but also impacts the predominant product types especially VFAs varieties [51]. As primary hydrolysate, VFAs can also serve as carbon sources for methanogenesis process. Under micro-aeration condition, VFAs can be converted into acetate predominantly, and acetate is a typical substrate for acetate-utilizing MPA to produce methane [61]. Table 2 presents the positive effects of micro-aeration based AD on FW hydrolysis and acidogenic fermentation. For example, micro-aeration system can promote 10 % FW hydrolysis efficiency than traditional AD system [60]. Additionally, according to previous studies, acetate and butyric are the two predominant hydrolysis products of carbohydrate-rich FW, while propionate is mainly derived from the degradation pathway of protein-rich substrates [9,75,76]. Micro-aeration technology increases the concentration of acetate and maintains total VFAs concentration at a low degree [61]. This merit alleviates negative effects of traditional FW AD process. As depicted in Fig. 2, different FW composition determines VFAs accumulation degree and micro-aeration based AD system at 120 L O_2/d aeration degree can relieve VFAs accumulation during carbohydrate-rich FW biodegradation. On the other hand, micro-aeration based AD system not only aggravates VFAs accumulation, but also lowers acetate production for lipid-rich FW disposal, which pose a challenge for lipid-rich FW biodegradation. Therefore, micro-aeration based AD system shows great potential to accelerate FW hydrolysis and maintain a low VFAs concentration without inhibition.

4.2. Regulating biogas production

In traditional AD system, methane is the most desirable economic product which is produced by MPA during complex substrate biodegradation process. Nevertheless, in FW biodegradation process, sulfur

Table 2
Effects of micro-aeration based AD on FW hydrolysis and acidogenic fermentation.

Bioreactor type and working volume	Micro-aeration rate or system ORP	Hydrolysis degree	Total VFAs concentration and predominant type	Reference
4.6 L LBR-10 L UASB	129, 258, 387 mL-air/g-TS d	129 mL-air/g-TS d: 60.1 %; 258 mL-air/g-TS d: 66.3 %; 387 mL-air/g-TS d: 78.1 %	129 mL-air/g-TS d: 0.26 g COD/g VS, 61 % acetate, 22 % butyric; 258 mL-air/g-TS d: 0.39 g COD/g VS, 58 % acetate, 32 % butyric; 387 mL-air/g-TS d: 0.19 g COD/g VS, 62 % acetate, 22 % butyric	[26]
0.5 L serum vial bottle	274 mL-air/g-TS d −100 mV to −200 mV	62.8 %	0.79 g COD/g VS; 58 % butyric, 30 % acetate	[51]
0.5 L serum vial bottle (Inoculate aerobic sludge)	274 mL-air/g-TS d 0 mV to −200 mV	60.1 %	0.53 g COD/g VS; 76 % butyric, 18 % acetate	[51]
0.2 L serum vial bottle	37.5 mL-O_2/L_R-d to pretreat FW −220 mV to −278 mV	10 % higher than anaerobic	7.44 g COD/L 37 % acetate, 25 % butyric	[60]
1.0–1.5 L CSTR	4.62, 7.69, 15.4 mL-air/L_R-min	–	4.62 mL-air/L_R-min: 0.11 g COD/g VS, 34 % isovalerate, 25 % isobutyrate; 7.69 mL-air/L_R-min: 0.98 g COD/g VS, 62 % acetate, 14 % isovalerate 15.4 mL-air/L_R-min: 0.74 g COD/g VS, 77 % acetate, 13 % isovalerate	[61]
20 L semi-pilot scale anaerobic hybrid biofilm reactor	400 L-O_2 to pretreat FW	29.7 %	6.75 g COD/L 78 % acetate, 12 % butyric	[67]

compounds in the substrate can be converted into hydrogen sulfide in biogas, which will reduce the purity of methane. Therefore, many physical, chemical and biological techniques have been applied to desulfurize biogas [77,78]. Among these technologies, dosing oxygen in anaerobic bioreactor is an effective and efficient technique to treat FW because it owns up to 99 % removal efficiency and bases on AD process intrinsically [79]. In general, the aeration rate and intensity need to be strictly controlled in micro-aeration based AD bioreactor because excess oxygen dosage reduces the activity of MPA to produce methane, and otherwise deficient oxygen dosage is incapable to show positive effects on methane yield [80]. Moreover, the composition of FW influences on micro-aeration based AD system in terms of methane production. Compared with traditional AD, cumulative methane production can increase during protein-rich FW micro-aeration treatment, while it decreases during carbohydrate-rich and lipid-rich FW micro-aeration treatment. Therefore, appropriate micro-aeration rate and FW composition are inspiring to be utilized throughout the entire AD process to enhance methane yield.

Table 3 records cumulative methane production in micro-aeration based AD system for FW treatment and compares the data with traditional AD system. In general, methane production under micro-aeration condition can not only increase but also decrease compared with traditional AD condition during FW biodegradation. As regards to FW pretreatment, compared with the no aeration condition, 5 mL O_2 can decrease 1 % of methane production, while 10, 15 and 20 mL O_2 can increase 8 %, 13 % and 12 %, respectively, which give a guidance about future aeration dosage aimed at increasing cumulative methane production [57].

On the other hand, it is obvious that micro-aeration based AD system may have negative effect on FW biomethane conversion if the oxygen dosage is controlled ineffectively [26]. Furthermore, micro-aeration condition plays a more important role in protein-rich FW treatment compared with lipid-rich FW and carbohydrate-rich FW treatment [26]. This can be ascribed to micro-aeration based AD can contribute to higher carbohydrates and proteins hydrolysis efficiency in seed sludge, while lipids hydrolysis efficiency shows no increase tendency without the addition of inoculum [54,81,82]. Therefore, micro-aeration exerts remarkable influence on biogas promotion ratio only if micro-aeration rate has been taken into sufficient consideration.

4.3. Enriching microbial species

Traditional AD process involves four steps including hydrolysis, acidogenesis, acetogenesis and methanogenesis and thus contributes to four species of functional microorganisms including hydrolytic bacterial, acidogens, acetogens and MPA. Among these four steps, methanogenesis is the most significant process because its performance is related to the ultimate products and the functional microbes can merely survive under strict anaerobic condition especially MPA. On the other hand, under micro-aeration environment, microbial species can be identified according to their tolerance on oxygen and thus they can comprise of strict and facultative anaerobes. In general, facultative microorganism plays the role as hydrolysis and acidogenic bacterial to consume the dissolved oxygen rapidly, and thus maintain an appropriate ORP value (lower than −300 mV) for MPA growth [83]. Meanwhile, in micro-aeration based AD system, different functional microorganisms can form microbial aggregates such as flocs, granules and biofilms which comprise of MPA core and bacteria surface to withstand adverse environment and promote syntrophic relationship as depicted in Fig. 2 [84,85]. Moreover, some species of MPA in the genera *Methanosarcina* and *Methanocella* have great potential to carry out methanogenesis activity under oxygen detrimental condition because they can encode various antioxidant enzymes such as superoxide reductase and rubredoxin oxidoreductase to resist oxygen stress and reduce reactive oxygen species detoxification [86,87]. As regards to FW treatment, the seed sludge containing a large quantity of facultative anaerobes such as species *Pseudomonas* in phylum *Firmicutes* shows great potential to promptly consume oxygen, and subsequently they can safeguard the secretion of additional extracellular hydrolases to accelerate FW hydrolysis [28,53].

Table 4 presents dominant functional microbes in micro-aeration based AD system for FW treatment and compares the result with traditional AD system. It is obvious that the relative abundance of phylum *Firmicutes* after micro-aeration based AD system increases over 10 % than in initial AD system apart from in Tang's research which enriches the abundance of MPA. Partial genus in phylum *Firmicutes* belong to facultative bacteria, which is known to produce cellulases, lipases, proteases and other extracellular enzymes to degrade FW [88]. Moreover, genus *Pseudomonas* in phylum *Firmicutes* has been proved to show great potential in degrading LCFAs under micro-aeration based AD system and this conclusion manifests a novel horizon for lipidic-rich FW

Table 3
Effects of micro-aeration based AD system on methane production regulation for FW treatment.

Working volume, bioreactor type and substrate characteristic	Micro-aeration rate or system ORP	Methane production rate	Increase or decrease percentage of methane production compared with AD	Reference
4.6 L LBR-10 L UASB, 7.9 g COD substrate	129, 258, 387 mL-air/g-TS d	129 mL-air/g-TS d: 0.17 L-methane/g-VS d; 258 mL-air/g-TS d: 0.27 L-methane/g-VS d; 387 mL-air/g-TS d: 0.19 L-methane/g-VS d	129 mL-air/g-TS d: decrease by 26 %; 258 mL-air/g-TS d: increase by 17 %; 387 mL-air/g-TS d: decrease by 17 %	[26]
3 L CSTR system, OLR: 5.15 ± 0.44 g-VS/L-d	7 mL-O_2/L_R-d, average ORP level: −100 mV	Around 0.4 L-methane/g-VS d	Increase by around 15 %	[27]
1.5 L PBR and 9 L UASB OLR: 2 g COD/L-d	254 mL-air/g-TS d	0.20 L-methane/g-VS d	–	[32]
1 L serum vail bottle, 5 g-VS/L (28 % carbohydrate-rich, 47 % protein-rich and 41 % lipid-rich)	120 L-air/d	Carbohydrate-rich: around 0.0037 L-methane/g-VS d; Protein-rich: 0.0059 L-methane/g-VS d; Lipid-rich: 0.004 L-methane/g-VS d	Carbohydrate-rich: increase by 6 % Protein-rich: increase by 45 % Lipid-rich: decrease by 9 %	[54]
0.2 L serum vial bottle $VS_{inoculum}/VS_{substrate}$ = 0.5	37.5 mL-O_2/L_R-d to pretreat FW −220 mV to −278 mV	0.012 L-methane/g-VS d	Increase by 23 %	[60]
0.2 L serum vial bottle	37.5 mL-O_2/L_R-d to pretreat FW −17 mV to −67 mV	0.01 L-methane/g-VS d	Increase by 16 %	[60]
20 L semi-pilot scale anaerobic hybrid biofilm reactor, OLR 10 g-COD/L-d	400 L-O_2 to pretreat FW	–	Increase by around 10 %	[67]
1 L SBR, $VS_{inoculum}/VS_{substrate}$ = 0.36	5 mL, 10 mL,15 mL 20 mL O_2 to pretreat FW	5 mL O_2: 0.0080 L-methane/g-VS d, 10 mL O_2: 0.0085 L-methane/g VS d, 15 mL O_2: 0.0089 L-methane/g VS d, 20 mL O_2: 0.0089 L-methane/g VS d	5 mL O_2: decrease by 1 % 10 mL O_2: increase by 8 % 15 mL O_2: increase by 13 % 20 mL O_2: increase by 12 %	[57]

Table 4
Effects of micro-aeration based AD system on primary functional microbe community for FW treatment.

Working volume, bioreactor type and temperature	Micro-aeration rate or system ORP	Primary microbial and their percentage in AD system	Primary microbial and their percentage in micro-aeration based AD system	Reference
3 L two-phase CSTR system, 35 °C	5 mL-O_2/L_R-d, average ORP level: −100 mV	58 % *Firmicutes*, 42 % *Proteobacteria*	72 % *Firmicutes*, 24 % *Bacteroidetes*, 2 % *Proteobacteria*, 2 % *Actinobacteria*	[27]
5 L gas circulation-type digester, 53 °C	7.5 % air of biogas-generation rate (v/v)	68 % *Firmicutes*, 28 % *Euryarchaeota*, 4 % *Thermotogae*	65 % *Firmicutes*, 35 % *Euryarchaeota*	[49]
500 mL serum vial bottle (Inoculate anaerobic sludge), 30 °C	274 mL-air/g-TS d −100 mV to −200 mV	40 % *Bacteroidetes*, 32.5 *Proteobacteria*, 21 % *Firmicutes*	48 % *Firmicutes*, 18 % *Proteobacteria*, 11 % *Bacteroidetes*	[51]
500 mL serum vial bottle (Inoculate aerobic sludge), 30 °C	274 mL-air/g-TS d 0 mV to −200 mV	40 % *Bacteroidetes*, 32.5 *Proteobacteria*, 21 % *Firmicutes*	42 % *Firmicutes*, 36 % *Proteobacteria*	[51]
1 L serum vial bottle, 37 °C	96 mL-O_2 to pretreat FW	52 % *Bacteroidetes*, 11 % *Synergistetes* 10 % *Firmicutes*	73 % *Firmicutes*, 5 % *Bacteroidetes*, 5 % *Synergistetes*	[53]
1.0–1.5 L CSTR	4.62 mL-air/L_R-min	38 % *Bacteroidetes*, 20 % *Firmicutes*	51 % *Firmicutes*, 11 % *Bacteroidetes*	[61]

treatment under modified anaerobic condition [28]. Furthermore, the relative abundance of phylum *Bacteroidetes* and *Proteobacteria* presents a downward tendency after micro-aeration system, especially as regards to the bioreactor aims at promoting hydrolysis and acidogenesis, the total abundance of phylum *Bacteroidetes* and *Proteobacteria* present a dramatic decrease by more than 15 % [27,61]. Though phylum *Bacteroidetes* and *Proteobacteria* are considered as the functional bacteria to conduct FW hydrolysis process, it shows inferior oxygen tolerance compared with *Firmicutes* and they could not consume oxygen during their metabolism process [89,90]. According to an deep-going analysis in terms of microbial species at class level, it is apparent that micro-aeration based AD system can enrich microbial class level compared with traditional AD, which is a symbol to maintain the stability of FW bioreactor [51].

In summary, micro-aeration based AD system can accelerate the diversity and activity of hydrolytic and fermentative microorganisms to decompose FW. These functional bacteria can further regulate and control total VFAs concentration and single VFAs concentration especially acetate concentration through the enhancement of their activity and metabolism, and thus it can maintain the stability of micro-aeration bioreactor for FW treatment.

5. Challenges and prospectives

Micro-aeration based AD system has shown great potential for FW treatment embodied in strengthening FW hydrolysis and acidogenic fermentation, regulating biogas production and enriching microbial species especially facultative microorganisms. However, there is still a lack of systematic research on FW biodegradation kinetic models and continuous-operation bioreactor for FW treatment under micro-aeration condition. Therefore, further studies are necessary to demonstrate FW micro-aeration digestion mechanism under different micro-aeration intensity and design stable pilot-scale micro-aeration bioreactor for FW treatment as follows.

(1) Effects of micro-aeration intensity and ORP value on FW micro-aeration digestion metabolism pathway and functional microbial community abundance deserve further research to clarify the mechanism behind micro-aeration based AD system to treat FW.
(2) Most researchers are confined to explore FW micro-aeration digestion process through serum vial bottles. Bioreactor optimization and amplification such as modified siphon-driven self-agitated anaerobic reactor needs further investigation under pilot-scale to preferably make comparison of traditional AD bioreactor and micro-aeration digestion bioreactor performance.
(3) Though kinetic modeling about biogas production and desulfurization with FW micro-aeration pretreatment has been put forward, systematic kinetic model such as modified ADM1-Ox under diverse micro-aeration intensities deserves to be developed according to the data not only from batch assays but also from long-term bioreactor performance.
(4) Effect of different micro-aeration strategies on FW biodegradation efficiency needs systematical investigation. Optimal micro-aeration strategy should be selected according to comprehensive factors such as bioreactor performance, FW characteristic and economic benefit.
(5) Oxygen dosing methods under micro-aeration based AD system for FW treatment cannot be restricted to conventional bubble aeration through aeration head, nanobubble aeration, membrane aeration, hydrogen peroxide decomposition and water electrolysis are also directions for future micro-aeration methods.

6. Conclusions

Micro-aeration based AD technology is a promising technology for FW treatment. Compared with traditional AD system, micro-aeration based AD system has a unique mechanism embodied in aerobic biodegradation pathway. Based on this, three common oxygen dosing strategies including micro-aeration during pretreatment phase, intermittent micro-aeration, persistent micro-aeration during whole experiment can be applied to create micro-aeration condition. By means of selecting the optimal oxygen dosing strategy and regulating the appropriate aeration dosage, Micro-aeration based AD technology can enhance bioreactor performance including strengthening FW hydrolysis and acidogenic fermentation, regulating biogas production and enriching microbial species. Nevertheless, due to incomplete systematic research in this field, challenges and gaps for future research about micro-aeration based AD system for FW treatment including optimized kinetic models and pilot-scale bioreactor.

CRediT authorship contribution statement

Xiang Li: Writing – original draft, Investigation, Formal analysis. **Yi-Juan Yan:** Visualization, Investigation, Data curation. **Chen-shun Lu:** Software, Data curation. **Hongyu Jiang:** Software, Investigation, Conceptualization. **Haiyuan Ma:** Writing – review & editing, Supervision. **Yong Hu:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

[1] C. Jin, S. Sun, D. Yang, W. Sheng, Y. Ma, W. He, G. Li, Anaerobic digestion: an alternative resource treatment option for food waste in China, Sci. Total Environ. 779 (2021) 146397, https://doi.org/10.1016/j.scitotenv.2021.146397.
[2] Y. Zhang, E. van Herpen, E.J. Van Loo, M. Pandelaere, M. Geuens, Save near-expired food: does a message to avoid food waste affect food purchase and household waste prevention behaviors? J. Clean. Prod. 384 (2023) 135555 https://doi.org/10.1016/j.jclepro.2022.135555.
[3] Q. Xu, S. Long, X. Liu, A. Duan, M. Du, Q. Lu, L. Leng, S.-Y. Leu, D. Wang, Insights into the occurrence, fate, impacts, and control of food additives in food waste anaerobic digestion: a review, Environ. Sci. Technol. 57 (2023) 6761–6775, https://doi.org/10.1021/acs.est.2c06345.
[4] A. Paranjpe, S. Saxena, P. Jain, A review on performance improvement of anaerobic digestion using co-digestion of food waste and sewage sludge, J. Environ. Manage. 338 (2023) 117733, https://doi.org/10.1016/j.jenvman.2023.117733.
[5] J. Ries, Z. Chen, Y. Park, Potential applications of food-waste-based anaerobic digestate for sustainable crop production practice, Sustainability 15 (2023) 8520, https://doi.org/10.3390/su15118520.
[6] Y.-J. Yan, X. Li, C.-S. Lu, T. Kobayashi, G.-Y. Zhen, Y. Hu, A review on start-up phase optimization of kitchen waste anaerobic digestion, Fermentation-Basel 9 (2023) 603, https://doi.org/10.3390/fermentation9070603.
[7] S. Zhang, C. Liang, M. Xiao, C. Chui, N. Wang, Y. Ji, Z. Wang, J. Shi, L. Liu, Metagenomic characterization of the enhanced performance of multicomponent synergistic thermophilic anaerobic co-digestion of food waste utilizing kitchen waste or garden waste as co-substrate, Water Res. 244 (2023) 120457, https://doi.org/10.1016/j.watres.2023.120457.
[8] K. Khanthong, R. Kadam, T. Kim, J. Park, Synergetic effects of anaerobic co-digestion of food waste and algae on biogas production, Bioresour. Technol. 382 (2023) 129208, https://doi.org/10.1016/j.biortech.2023.129208.
[9] Q. Wang, G. Zhang, L. Chen, N. Yang, Y. Wu, W. Fang, R. Zhang, X. Wang, C. Fu, P. Zhang, Volatile fatty acid production in anaerobic fermentation of food waste saccharified residue: effect of substrate concentration, Waste Manag. 164 (2023) 29–36, https://doi.org/10.1016/j.wasman.2023.03.045.
[10] C. Mao, Y. Feng, X. Wang, G. Ren, Review on research achievements of biogas from anaerobic digestion, Renew. Sustain. Energy Rev. 45 (2015) 540–555, https://doi.org/10.1016/j.rser.2015.02.032.
[11] W. Guo, D. Li, Z. Zhang, R. Mo, Y. Peng, Y. Li, A novel approach for the fractionation of organic components and microbial degraders in ADM1 and model validation based on the methanogenic potential, Water Res. 236 (2023) 119945, https://doi.org/10.1016/j.watres.2023.119945.
[12] H. Ding, M.A. Barlaz, F.L. de los Reyes III, D.F. Call, Influence of inoculum type on volatile fatty acid and methane production in short-term anaerobic food waste digestion tests, ACS Sustain. Chem. Eng. 10 (2022) 17071–17080, https://doi.org/10.1021/acssuschemeng.2c04080.
[13] D. Zhang, Y. Wei, S. Wu, L. Zhou, Consolidation of hydrogenotrophic methanogenesis by sulfidated nanoscale zero-valent iron in the anaerobic digestion of food waste upon ammonia stress, Sci. Total Environ. 822 (2022) 153531, https://doi.org/10.1016/j.scitotenv.2022.153531.
[14] S. Zhang, M. Xiao, C. Liang, C. Chui, N. Wang, J. Shi, L. Liu, Multivariate insights into enhanced biogas production in thermophilic dry anaerobic co-digestion of food waste with kitchen waste or garden waste: process properties, microbial

communities and metagenomic analyses, Bioresour. Technol. 361 (2022) 127684, https://doi.org/10.1016/j.biortech.2022.127684.
[15] J. Zhao, Z. Wang, J. Li, B. Yan, G. Chen, Pyrolysis of food waste and food waste solid digestate: a comparative investigation, Bioresour. Technol. 354 (2022) 127191, https://doi.org/10.1016/j.biortech.2022.127191.
[16] Z. Huang, Q. Niu, W. Nie, X. Li, C. Yang, Effects of heavy metals and antibiotics on performances and mechanisms of anaerobic digestion, Bioresour. Technol. 361 (2022) 127683, https://doi.org/10.1016/j.biortech.2022.127683.
[17] O. Golovko, L. Ahrens, J. Schelin, M. Sorengard, K.-J. Bergstrand, H. Asp, M. Hultberg, K. Wiberg, Organic micropollutants, heavy metals and pathogens in anaerobic digestate based on food waste, J. Environ. Manage. 313 (2022) 114997, https://doi.org/10.1016/j.jenvman.2022.114997.
[18] D.-H. Wang, S.-J. Lian, R.-N. Wang, H. Zou, R.-B. Guo, S.-F. Fu, Enhanced anaerobic digestion of food waste by metal cations and mechanisms analysis, Renew. Energy 218 (2023) 119386, https://doi.org/10.1016/j.renene.2023.119386.
[19] Z. Zhou, Q. Ming, Y. An, D. Ruan, G. Chen, H. Wei, M. Wang, Z. Wu, Performance and microbial community analysis of anaerobic sludge digestion enhanced by in-situ microaeration, J. Water Process. Eng. 42 (2021) 102171, https://doi.org/10.1016/j.jwpe.2021.102171.
[20] Z. Zhang, Y. Yu, H. Xi, Y. Zhou, Review of micro-aeration hydrolysis acidification for the pretreatment of toxic and refractory organic wastewater, J. Clean. Prod. 317 (2021) 128343, https://doi.org/10.1016/j.jclepro.2021.128343.
[21] Q. Cao, W. Zhang, T. Lian, S. Wang, F. Yin, T. Zhou, X. Wei, H. Dong, Revealing mechanism of micro-aeration for enhancing volatile fatty acids production from swine manure, Bioresour. Technol. 365 (2022) 128140, https://doi.org/10.1016/j.biortech.2022.128140.
[22] N. Duc, S.K. Khanal, A little breath of fresh air into an anaerobic system: how microaeration facilitates anaerobic digestion process, Biotechnol. Adv. 36 (2018) 1971–1983, https://doi.org/10.1016/j.biotechadv.2018.08.007.
[23] Q. Chen, W. Wu, D. Qi, Y. Ding, Z. Zhao, Review on microaeration-based anaerobic digestion: state of the art, challenges, and prospectives, Sci. Total Environ. 710 (2020) 136388, https://doi.org/10.1016/j.scitotenv.2019.136388.
[24] D. Botheju, B. Lie, R. Bakke, Oxygen effects in anaerobic digestion, Model. Identif. Control 30 (2009) 191–201, https://doi.org/10.4173/mic.2009.4.1.
[25] N.J.R. Kraakman, I. Diaz, M. Fdz-Polanco, R. Munoz, Large-scale micro-aerobic digestion studies at municipal water resource recovery facilities for process-integrated biogas desulfurization, J. Water Process. Eng. 53 (2023) 103643, https://doi.org/10.1016/j.jwpe.2023.103643.
[26] S. Xu, A. Selvam, J.W.C. Wong, Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste, Waste Manag. 34 (2014) 363–369, https://doi.org/10.1016/j.wasman.2013.10.038.
[27] J.W. Lim, J.A. Chiam, J.-Y. Wang, Microbial community structure reveals how microaeration improves fermentation during anaerobic co-digestion of brown water and food waste, Bioresour. Technol. 171 (2014) 132–138, https://doi.org/10.1016/j.biortech.2014.08.050.
[28] M. Salome Duarte, S.A. Silva, A.F. Salvador, A.J. Cavaleiro, A.J.M. Stams, M. Madalena Alves, M. Alcina Pereira, Insight into the role of facultative bacteria stimulated by microaeration in continuous bioreactors converting LCFA to methane, Environ. Sci. Technol. 52 (2018) 6497–6507, https://doi.org/10.1021/acs.est.8b00894.
[29] F. Zan, A. Iqbal, G. Guo, X. Liu, J. Dai, G.A. Ekama, G. Chen, Integrated food waste management with wastewater treatment in Hong Kong: transformation, energy balance and economic analysis, Water Res. 184 (2020) 116155, https://doi.org/10.1016/j.watres.2020.116155.
[30] C. Macintosh, S. Astals, C. Sembera, A. Ertl, J.E. Drewes, P.D. Jensen, K. Koch, Successful strategies for increasing energy self-sufficiency at Gruneck wastewater treatment plant in Germany by food waste co-digestion and improved aeration, Appl. Energy 242 (2019) 797–808, https://doi.org/10.1016/j.apenergy.2019.03.126.
[31] A. Ballo, P. Czetany, K.S. Busznyakne, L. Mark, N. Mike, A. Torok, A. Szanto, G. Mate, Oxido-reduction potential as a method to determine oxidative stress in semen samples, Int. J. Mol. Sci. 24 (2023) 11981, https://doi.org/10.3390/ijms241511981.
[32] F. Canul Bacab, E. Espana Gamboa, J.E. Ruiz Espinoza, R.M. Leal-Bautista, R. Tapia Tussell, J. Dominguez Maldonado, B. Canto Canche, L. Alzate-Gaviria, Two phase anaerobic digestion system of municipal solid waste by utilizing microaeration and granular activated carbon, Energies 13 (2020) 933, https://doi.org/10.3390/en13040933.
[33] H. Luo, D. Zhang, M. Taylor, C. Nguyen, Z.-W. Wang, Aeration in sludge holding tanks as an economical means for biosolids odor control-a case study, Water Environ. Res. 93 (2021) 1808–1818, https://doi.org/10.1002/wer.1582.
[34] Y.F. Zhang, J.Z. Li, J. Meng, K. Sun, H. Yan, A neutral red mediated electro-fermentation system of *Clostridium beijerinckii* for effective co-production of butanol and hydrogen, Bioresour. Technol. 332 (2021) 125097, https://doi.org/10.1016/j.biortech.2021.125097.
[35] R. Ni, C. Xu, X. Shi, S. Yang, L. Li, X. Peng, L. Song, Acetoclastic methanogenesis pathway stability despite the high microbial taxonomic variability in the transition from acidogenesis to methanogenesis during food waste anaerobic digestion, J. Clean. Prod. 372 (2022) 133758, https://doi.org/10.1016/j.jclepro.2022.133758.
[36] P. Yang, Y. Peng, H. Liu, D. Wu, R. Yuan, X. Wang, L. Li, X. Peng, Multi-scale analysis of the foaming mechanism in anaerobic digestion of food waste: from physicochemical parameter, microbial community to metabolite response, Water Res. 218 (2022) 118482, https://doi.org/10.1016/j.watres.2022.118482.
[37] L. Ding, Y. Chen, Y. Xu, B. Hu, Improving treatment capacity and process stability via a two-stage anaerobic digestion of food waste combining solid-state acidogenesis and leachate methanogenesis/recirculation, J. Clean. Prod. 279 (2021) 123644, https://doi.org/10.1016/j.jclepro.2020.123644.
[38] X. Zhou, T. Liu, S. Zhang, B. Kang, X. Duan, Y. Yan, L. Feng, Y. Chen, Metagenomic insight of fluorene-boosted sludge acidogenic fermentation: metabolic transformation of amino acids and monosaccharides, Sci. Total Environ. 865 (2023) 161122, https://doi.org/10.1016/j.scitotenv.2022.161122.
[39] P. Photiou, I. Vyrides, Calcined eggshells in anaerobic digestion: buffering acidification in AD and evaluating end products from phosphate adsorption as soil conditioners, J. Environ. Chem. Eng. 10 (2022) 107957, https://doi.org/10.1016/j.jece.2022.107957.
[40] C. Morales-Polo, M.D.M. Cledera-Castro, M. Revuelta-Aramburu, K. Hueso-Kortekaas, Anaerobic digestion of organic fraction combinations from food waste, for an optimal dynamic release of biogas, using H_2 as an indicator, Sci. Total Environ. 857 (2023) 159727, https://doi.org/10.1016/j.scitotenv.2022.159727.
[41] M. Ye, Y.Y. Li, Methanogenic treatment of dairy wastewater: a review of current obstacles and new technological perspectives, Sci. Total Environ. 866 (2023) 161447, https://doi.org/10.1016/j.scitotenv.2023.161447.
[42] X. Li, Y. Yang, C.-S. Lu, T. Kobayashi, Z. Kong, Y. Hu, Oleate impacts on acetoclastic and hydrogenotrophic methanogenesis under mesophilic and thermophilic conditions, Int. J. Environ. Res. Public Health 20 (2023) 3423, https://doi.org/10.3390/ijerph20043423.
[43] Y. Liu, Q. Xiao, Z. Jia, C. Wang, X. Ye, J. Du, X. Kong, Y. Xi, Relieving ammonia nitrogen inhibition in high concentration anaerobic digestion of rural organic household waste by Prussian blue analogue nanoparticles addition, Bioresour. Technol. 330 (2021) 124979, https://doi.org/10.1016/j.biortech.2021.124979.
[44] C. Liu, W. Wang, N. Anwar, Z. Ma, G. Liu, R. Zhang, Effect of organic loading rate on anaerobic digestion of food waste under mesophilic and thermophilic conditions, Energy Fuel 31 (2017) 2976–2984, https://doi.org/10.1021/acs.energyfuels.7b00018.
[45] S. Mothe, V.R. Polisetty, Review on anaerobic digestion of rice straw for biogas production, Environ. Sci. Pollut. Res. 28 (2021) 24455–24469, https://doi.org/10.1007/s11356-020-08762-9.
[46] Y. Hu, H. Ma, C. Shi, T. Kobayashi, K.-Q. Xu, Nutrient augmentation enhances biogas production from sorghum mono-digestion, Waste Manag. 119 (2021) 63–71, https://doi.org/10.1016/j.wasman.2020.09.041.
[47] G.Z. Kyzas, G. Bomis, R.I. Kosheleva, E.K. Efthimiadou, E.P. Favvasa, M. Kostoglou, A.C. Mitropoulos, Nanobubbles effect on heavy metal ions adsorption by activated carbon, Chem. Eng. J. 356 (2019) 91–97, https://doi.org/10.1016/j.cej.2018.09.019.
[48] R. Zhu, D.-H. Wang, Y. Zheng, H. Zou, S.-F. Fu, Understanding the mechanisms behind micro-aeration to enhance anaerobic digestion of corn straw, Fuel 318 (2022) 123604, https://doi.org/10.1016/j.fuel.2022.123604.
[49] Y.Q. Tang, T. Shigematsu, S. Ikbal, K. Kida Morimura, The effects of micro-aeration on the phylogenetic diversity of microorganisms in a thermophilic anaerobic municipal solid-waste digester, Water Res. 38 (2004) 2537–2550, https://doi.org/10.1016/j.watres.2004.03.012.
[50] A. Mahdy, Y. Song, A. Salama, W. Qiao, R. Dong, Simultaneous H₂S mitigation and methanization enhancement of chicken manure through the introduction of the micro-aeration approach, Chemosphere 253 (2020) 126687, https://doi.org/10.1016/j.chemosphere.2020.126687.
[51] J. Yin, X. Yu, Y. Zhang, D. Shen, M. Wang, Y. Long, T. Chen, Enhancement of acidogenic fermentation for volatile fatty acid production from food waste: effect of redox potential and inoculum, Bioresour. Technol. 216 (2016) 996–1003, https://doi.org/10.1016/j.biortech.2016.06.053.
[52] T.I.M. Grootscholten, D.P.B.T.B. Strik, K.J.J. Steinbusch, C.J.N. Buisman, H.V. M. Hamelers, Two-stage medium chain fatty acid (MCFA) production from municipal solid waste and ethanol, Appl. Energy 116 (2014) 223–229, https://doi.org/10.1016/j.apenergy.2013.11.061.
[53] S.-F. Fu, X.-S. Shi, M. Dai, R.-B. Guo, Effect of different mixed microflora on the performance of thermophilic microaerobic pretreatment, Energy Fuel 30 (2016) 6413–6418, https://doi.org/10.1021/acs.energyfuels.6b00440.
[54] R. Rafieenia, F. Girotto, W. Peng, R. Cossu, A. Pivato, R. Raga, M.C. Lavagnolo, Effect of aerobic pre-treatment on hydrogen and methane production in a two-stage anaerobic digestion process using food waste with different compositions, Waste Manag. 59 (2017) 194–199, https://doi.org/10.1016/j.wasman.2016.10.028.
[55] C. Wu, Y. Li, Y. Zhou, Z. Li, S. Zhang, H. Liu, Upgrading the Chinese biggest petrochemical wastewater treatment plant: technologies research and full scale application, Sci. Total Environ. 633 (2018) 189–197, https://doi.org/10.1016/j.scitotenv.2018.03.164.
[56] Z. Lu, J.A. Imlay, When anaerobes encounter oxygen: mechanisms of oxygen toxicity, tolerance and defence, Nat. Rev. Microbiol. 19 (2021) 774–785, https://doi.org/10.1038/s41579-021-00583-y.
[57] X. Zhen, X. Zhang, S. Li, M. Li, J. Kang, Effect of micro-oxygen pretreatment on gas production characteristics of anaerobic digestion of kitchen waste, J. Mater. Cycles Waste Manage. 22 (2020) 1852–1858, https://doi.org/10.1007/s10163-020-01072-9.
[58] L. Feng, X. Li, S.Y. Gu, Y. Li, M. Zhang, Y.S. Yan, W.Y. Zhang, Kinetic modelling of gas generated from anaerobic fermentation of kitchen waste with microaerobic pretreatment, Appl. Ecol. Environ. Res. 17 (2019) 10671–10683, https://doi.org/10.15666/aeer/1705_1067110683.
[59] O. Khongsumran, P. Intanoo, P. Rangsunvigit, S. Chavadej, M.J.C.E. T. Leethochawalit, Enhancement of anaerobic digestion of cellulosic fraction in

cassava production wastewater by microaeration, chemical, Eng. Trans. 39 (2014) 553–558, https://doi.org/10.3303/CET1439093.
[60] J.W. Lim, J.-Y. Wang, Enhanced hydrolysis and methane yield by applying microaeration pretreatment to the anaerobic co-digestion of brown water and food waste, Waste Manag. 33 (2013) 813–819, https://doi.org/10.1016/j.wasman.2012.11.013.
[61] Y. Wen, H. Jiang, R. Qian, S. Liu, X. Tang, W. Huang, H. Chen, Micro-aerated fermentation enhances acetate production from high-rate activated sludge to supply carbon source for heterotrophic denitratation, Chem. Eng. J. 446 (2022) 136894, https://doi.org/10.1016/j.cej.2022.136894.
[62] W.-Y. Sean, Y.-Y. Chu, L.L. Mallu, J.-G. Chen, H.-Y. Liu, Energy consumption analysis in wastewater treatment plants using simulation and SCADA system: case study in northern Taiwan, J. Clean. Prod. 276 (2020) 124248, https://doi.org/10.1016/j.jclepro.2020.124248.
[63] L.D. Nghiem, P. Manassa, M. Dawson, S.K. Fitzgerald, Oxidation reduction potential as a parameter to regulate micro-oxygen injection into anaerobic digester for reducing hydrogen sulphide concentration in biogas, Bioresour. Technol. 173 (2014) 443–447, https://doi.org/10.1016/j.biortech.2014.09.052.
[64] F. Garcia-Ochoa, E. Gomez, Bioreactor scale-up and oxygen transfer rate in microbial processes: an overview, Biotechnol. Adv. 27 (2009) 153–176, https://doi.org/10.1016/j.biotechadv.2008.10.006.
[65] I. Diaz, I. Ramos, M. Fdz-Polanco, Economic analysis of microaerobic removal of H_2S from biogas in full-scale sludge digesters, Bioresour. Technol. 192 (2015) 280–286, https://doi.org/10.1016/j.biortech.2015.05.048.
[66] J.P.S. Siqueira, A.M. Pereira, A.M.M. Dutra, P.I.M. Firmino, A.B. dos Santos, Process bioengineering applied to BTEX degradation in microaerobic treatment systems, J. Environ. Manage. 223 (2018) 426–432, https://doi.org/10.1016/j.jenvman.2018.06.066.
[67] O. Sarkar, S.V. Mohan, Pre-aeration of food waste to augment acidogenic process at higher organic load: valorizing biohydrogen, volatile fatty acids and biohythane, Bioresour. Technol. 242 (2017) 68–76, https://doi.org/10.1016/j.biortech.2017.05.053.
[68] X. Fei, C. Zhou, Study on the enzymatic hydrolysis performances of carbohydrates, proteins and oils after γ-ray irradiation, J. Radioanal. Nucl. Chem. 332 (2023) 2213–2219, https://doi.org/10.1007/s10967-023-08856-x.
[69] S. Rehman, Y.S. Yang, R.D. Patria, T. Zulfiqar, N.K. Khanzada, R.J. Khan, C.S. K. Lin, D.-J. Lee, S.-Y. Leu, Substrate-related factors and kinetic studies of carbohydrate-rich food wastes on enzymatic saccharification, Bioresour. Technol. 390 (2023) 129858, https://doi.org/10.1016/j.biortech.2023.129858.
[70] D. Zhao, B. Yan, C. Liu, B. Yao, L. Luo, Y. Yang, L. Liu, F. Wu, Y. Zhou, Mitigation of acidogenic product inhibition and elevated mass transfer by biochar during anaerobic digestion of food waste, Bioresour. Technol. 338 (2021) 125531, https://doi.org/10.1016/j.biortech.2021.125531.
[71] H. Pang, Y. Xu, R. Ren, J. He, X. Pan, L. Wang, Enhanced anaerobic digestion of waste activated sludge by alkaline protease-catalyzing hydrolysis: role and significance of initial pH adjustment, Chem. Eng. J. 467 (2023) 143323, https://doi.org/10.1016/j.cej.2023.143323.
[72] Preethi, R.J. Banu, S. Varjani, P. Sivashanmugam, V.K. Tyagi, M. Gunasekaran, Breakthrough in hydrolysis of waste biomass by physico-chemical pretreatment processes for efficient anaerobic digestion, Chemosphere 294 (2022) 133617, https://doi.org/10.1016/j.chemosphere.2022.133617.
[73] X. Yang, L. Zhao, X. Zhao, P. Wang, Y. Zheng, L. Ren, Effect and mechanism analysis of dual enzyme pretreatment on enhancement of volatile fatty acids production in kitchen waste anaerobic acidification, J. Environ. Chem. Eng. 11 (2023) 110109, https://doi.org/10.1016/j.jece.2023.110109.
[74] S. Manikandan, S. Vickram, R. Sirohi, R. Subbaiya, R.Y. Krishnan, N. Karmegam, C. Sumathijones, R. Rajagopal, S.W. Chang, B. Ravindran, M.K. Awasthi, Critical review of biochemical pathways to transformation of waste and biomass into bioenergy, Bioresour. Technol. 372 (2023) 128679, https://doi.org/10.1016/j.biortech.2023.128679.
[75] W. Fang, X. Zhang, P. Zhang, J. Wan, H. Guo, D.S.M. Ghasimi, X. Carol Morera, T. Zhang, Overview of key operation factors and strategies for improving fermentative volatile fatty acid production and product regulation from sewage sludge, J. Environ. Sci. 87 (2020) 93–111, https://doi.org/10.1016/j.jes.2019.05.027.
[76] K. Dudek, G. Buitron, I. Valdez-Vazquez, Nutrient influence on acidogenesis and native microbial community of *Agave* bagasse, Ind. Crop. Prod. 170 (2021) 113751, https://doi.org/10.1016/j.indcrop.2021.113751.
[77] G.K. Hassan, W.H. Mahmoud, A. Al-sayed, S.H. Ismail, A.A. El-Sherif, S.M. Abd El Wahab, Multi-functional of TiO_2@Ag core-shell nanostructure to prevent hydrogen sulfide formation during anaerobic digestion of sewage sludge with boosting of bio-CH_4 production, Fuel 333 (2023) 126608, https://doi.org/10.1016/j.fuel.2022.126608.
[78] F. Egger, T. Hulsen, D.J. Batstone, Continuous H_2S removal from biogas using purple phototrophic bacteria, Chem. Eng. J. 454 (2023) 140449, https://doi.org/10.1016/j.cej.2022.140449.
[79] C.-Y. Lau, J. Guan, H.-Y. Tse, C.S. Yeung, S.-Y. Leu, Smart control for micro-aeration toward sulfide removal in sewer systems at low costs, Desalin. Water Treat. 200 (2020) 369–375, https://doi.org/10.5004/dwt.2020.26117.
[80] B. Guo, N. Yu, D.G. Weissbrodt, Y. Liu, Effects of micro-aeration on microbial niches and antimicrobial resistances in blackwater anaerobic digesters, Water Res. 196 (2021) 117035, https://doi.org/10.1016/j.watres.2021.117035.
[81] K. Liu, L. Lv, W. Li, X. Wang, M. Han, Z. Ren, W. Gao, P. Wang, X. Liu, L. Sun, G. Zhang, Micro-aeration and leachate recirculation for the acceleration of landfill stabilization: enhanced hydrolytic acidification by facultative bacteria, Bioresour. Technol. 387 (2023) 129615, https://doi.org/10.1016/j.biortech.2023.129615.
[82] Y. Wang, H. Wang, J. Zhu, H. Chen, Review on enhanced hydrolysis acidification strategies for pre-treatment of refractory wastewater or VFAs production: mechanisms and application, J. Clean. Prod. 410 (2023) 137252, https://doi.org/10.1016/j.jclepro.2023.137252.
[83] M.L.U. Rehman, A. Iqbal, C.-C. Chang, W. Li, M. Ju, Anaerobic digestion, Water Environ. Res. 91 (2019) 1253–1271, https://doi.org/10.1002/wer.1219.
[84] M. Layer, A. Brison, M.G. Villodres, M. Staehle, F. Hazi, I. Takacs, E. Morgenroth, N. Derlon, Microbial conversion pathways of particulate organic substrate conversion in aerobic granular sludge systems: limited anaerobic conversion and the essential role of flocs, Environ. Sci. Water Res. Technol. 8 (2022) 1236–1251, https://doi.org/10.1039/d1ew00841b.
[85] T. Niu, Z. Zhou, X. Shen, W. Qiao, L.-M. Jiang, W. Pan, J. Zhou, Effects of dissolved oxygen on performance and microbial community structure in a micro-aerobic hydrolysis sludge in *situ* reduction process, Water Res. 90 (2016) 369–377, https://doi.org/10.1016/j.watres.2015.12.050.
[86] Z. Lyu, Y. Lu, Metabolic shift at the class level sheds light on adaptation of methanogens to oxidative environments, ISME J. 12 (2018) 411–423, https://doi.org/10.1038/ismej.2017.173.
[87] N. Yu, B. Guo, Y. Zhang, L. Zhang, Y. Zhou, Y. Liu, Different micro-aeration rates facilitate production of different end-products from source-diverted blackwater, Water Res. 177 (2020) 115783, https://doi.org/10.1016/j.watres.2020.115783.
[88] L. Leven, A.R.B. Eriksson, A. Schnurer, Effect of process temperature on bacterial and archaeal communities in two methanogenic bioreactors treating organic household waste, FEMS Microbiol. Ecol. 59 (2007) 683–693, https://doi.org/10.1111/j.1574-6941.2006.00263.x.
[89] W. Li, Y. Liu, B. Wu, L. Gu, R. Deng, Upgrade the high-load anaerobic digestion and relieve acid stress through the strategy of side-stream micro-aeration: biochemical performances, microbial response and intrinsic mechanisms, Water Res. 221 (2022) 118850, https://doi.org/10.1016/j.watres.2022.118850.
[90] L. Zhu, B. Wu, Y. Liu, J. Zhang, R. Deng, L. Gu, Strategy to enhance semi-continuous anaerobic digestion of food waste by combined use of calcium peroxide and magnetite, Water Res. 221 (2022) 118801, https://doi.org/10.1016/j.watres.2022.118801.